



Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)



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WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10212CS292 / Applied Programming Skills

Unit IV: Network and Relationship-Driven Data Structures Course Outcome: Utilize graph and set-based structures to analyze connectivity, traversal, and relationship-driven problems				
Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 7	Session 31	Graph	Task 54: Find if Path Exists in Graph Task 55: Find the Town Judge	Leet Code
	Session 32	Breadth First Search	Task 56: Rotting Oranges	
	Session 33	Depth First Search	Task 57: Number of Provinces Task 58: Max Area of Island	
	Session 34	Connected components	Task 59: Keys and Rooms	
	Session 35	Cycle detection	Task 60: Course Schedule Task 61: Course Schedule II	

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 8	Session 36	Topological sorting	Task 62: Sort Items by Groups Respecting Dependencies	takeuforward / Leet Code
	Session 37	Shortest path in unweighted graphs	Task 63: Shortest Path with Alternating Colors Task 64: 01 Matrix	
	Session 38	Grid traversal (islands, flood fill)	Task 65: Number of Islands Task 66: Flood Fill	
	Session 39	Union Find (Disjoint Set Union)	Task 67: Accounts Merge	
	Session 40	HashSet patterns – Frequency map and sorting by keys/values	Task 68: Happy Number Task 69: Unique Email Addresses Task 70: Group Anagrams	